



CardioTriage AI

Machine Learning Phase — Detailed Walkthrough

From raw cardiac records to a deployed risk-and-phenotype engine (Steps A1–A8)

920

patients

3

models compared

0.91

ROC-AUC

0.96

recall (tuned)

Where the Machine Learning Phase Fits

1 • Data & ML

ML risk engine — predicts per-patient risk probability + phenotype

2 • Classical AI core

A* search · CSP allocator · forward-chaining knowledge base

3 • Orchestration

Gemini + LangChain agent — decides which tool to call & explains

4 • Presentation

Interactive web triage board (ranking, paths, allocations)

This deck covers Layer 1

Phase A builds the ML risk engine — Steps A1–A8.

Its outputs (risk probability + phenotype) become the inputs every layer above consumes.

The LLM never makes a medical call alone — it coordinates deterministic tools.

The Dataset — UCI Heart Disease (combined)

- 920 patients × 16 clinical columns.
- Pooled from four hospitals (multi-site).
- Target 'num' (0–4) → binarized: 0 = no disease, 1–4 = disease.
- Contains real missing values — handled honestly, not dropped.

304

Cleveland records

293

Hungary records

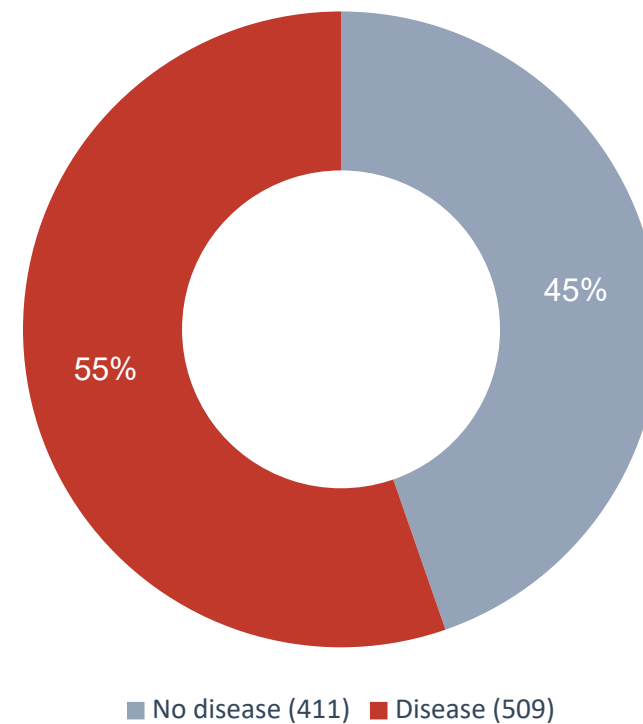
200

VA Long Beach records

123

Switzerland records

Outcome balance



Eight Steps, One Pipeline

A1 **Inspect**

shape, types, missingness

A2 **Preprocess**

clean · encode · scale · split

A3 **Explore**

EDA: what separates sick

A4 **Baseline**

Logistic Regression

A5 **Compare**

RF & Gradient Boosting

A6 **Tune**

bias-variance · reg · threshold

A7 **Cluster**

K-means risk phenotypes

A8 **Save**

persist + bridge to AI

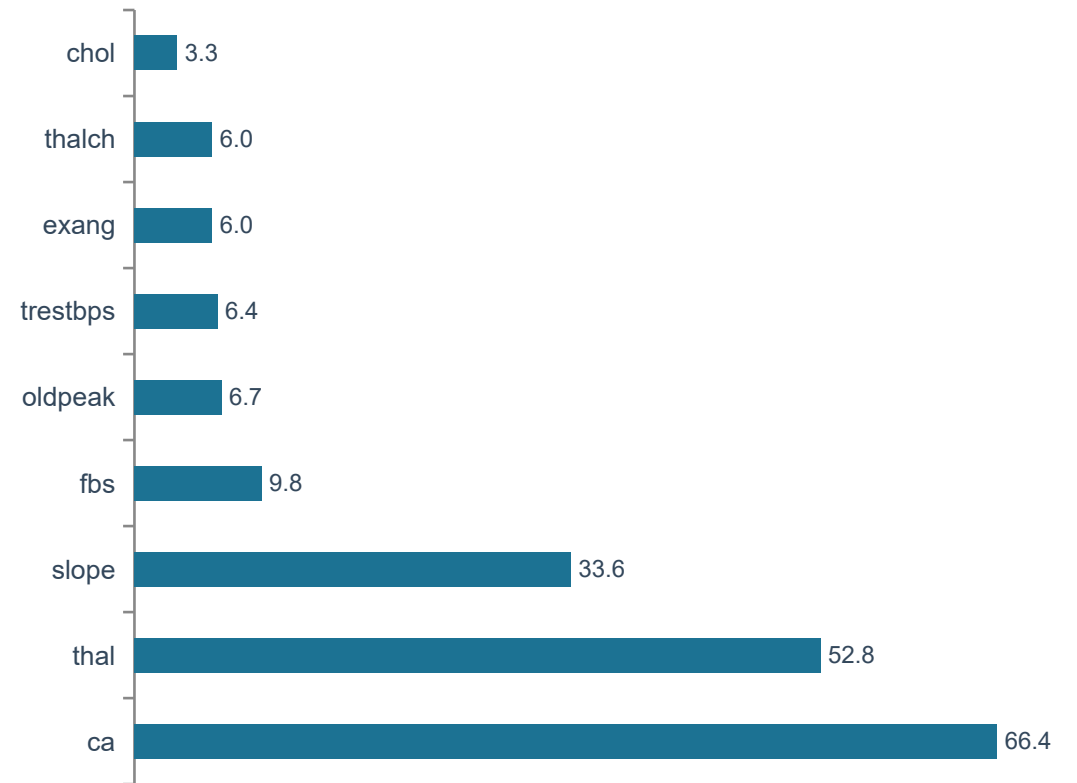
Load & Inspect the Data

- Read the CSV into a pandas DataFrame — change nothing yet.
- Confirmed 920 rows × 16 columns; identified each column's meaning & type.
- Target distribution: 411 healthy vs 509 diseased (after binarizing).
- Surfaced the missing-value map (right) — the central challenge.

Key finding

ca (66%) and thal (53%) are missing for over half of patients — too valuable to drop, too sparse to ignore. This drove the A2 strategy.

Missing values by column (%)



Preprocessing — Leakage-Safe Pipeline

- Binarize target: num 1–4 → 1 (disease present).
- Drop id (row number) and dataset (hospital — a confound).
- Fix hidden missingness: chol = 0 is impossible → 172 cases set to missing.
- Add ca_missing / thal_missing flags so the model knows a value was estimated.
- Median/mode impute · one-hot encode · standardize — all inside one Pipeline.
- Fit on TRAIN only, apply to TEST → no information leaks across the split.

736 / 184

stratified train / test

0.553 / 0.554

disease rate · train vs test

15 → 27

features after encoding

0

missing values remaining

Why leakage matters

If fill-values or scales were learned from the whole dataset, the test score would be inflated and dishonest.

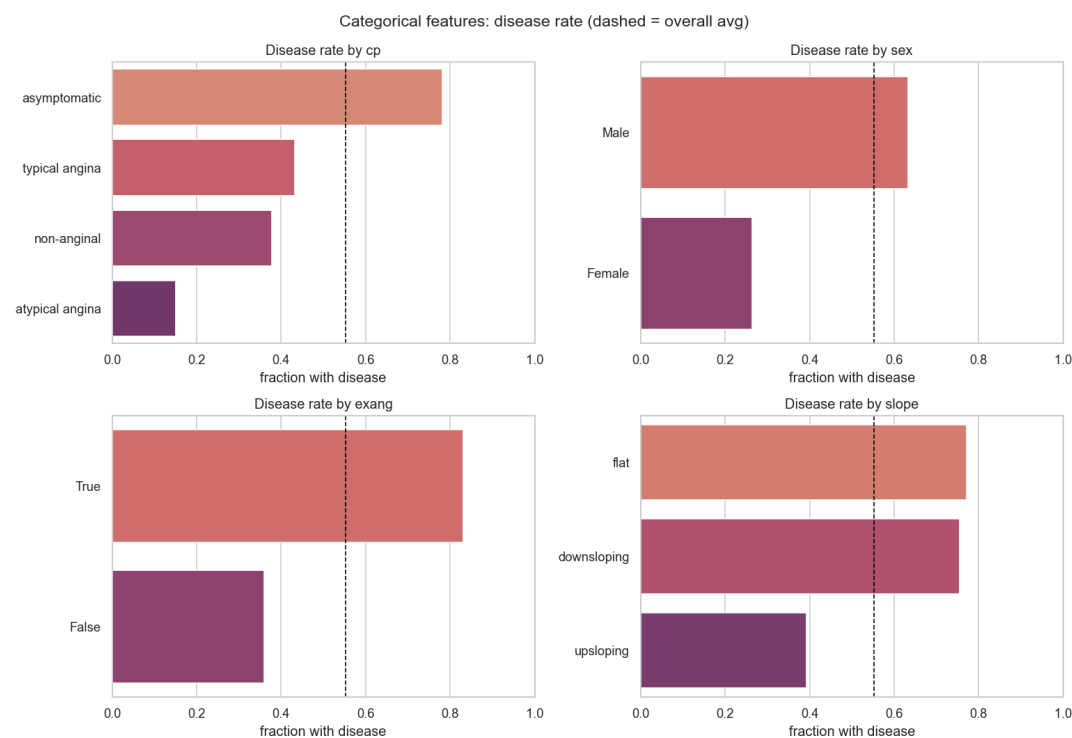
Exploratory Analysis — What Separates Sick from Healthy

Mean value by outcome (training set)

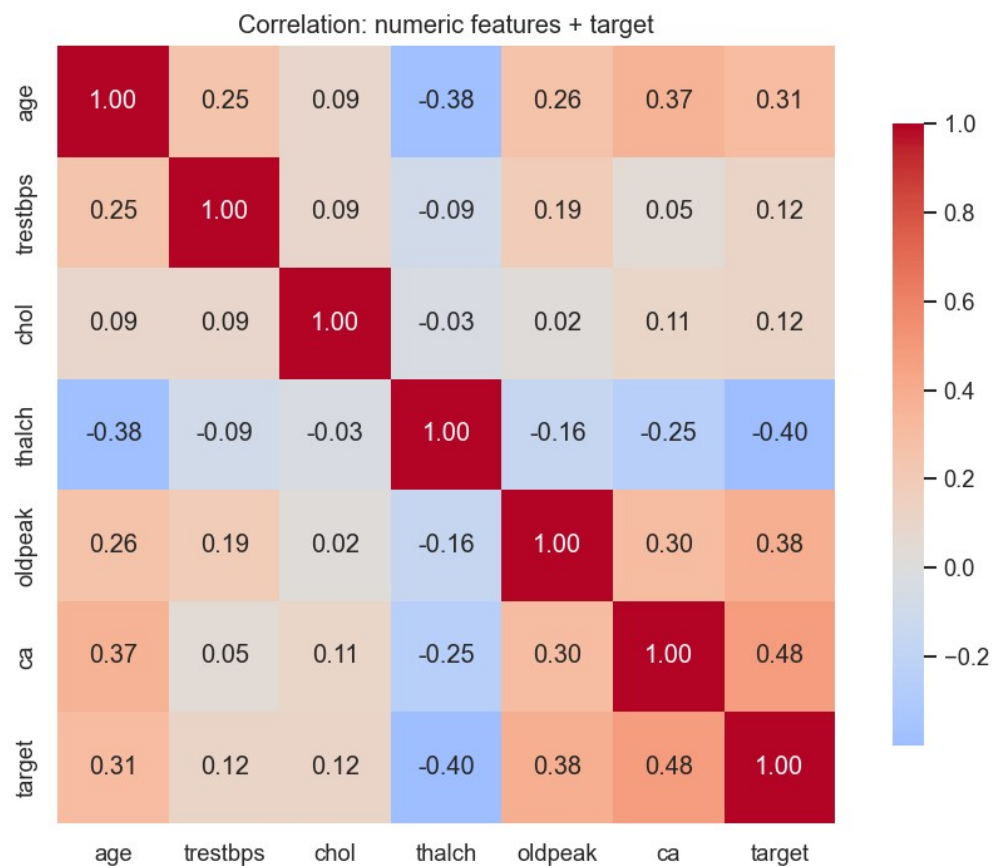
Feature	Disease	Healthy	Signal
thalch (max HR)	128.2	149.1	strong
oldpeak	1.3	0.4	strong
ca (vessels)	1.1	0.2	strong
age	56.2	50.4	moderate
chol	254	240	weak
trestbps	134	130	weak

Silent danger

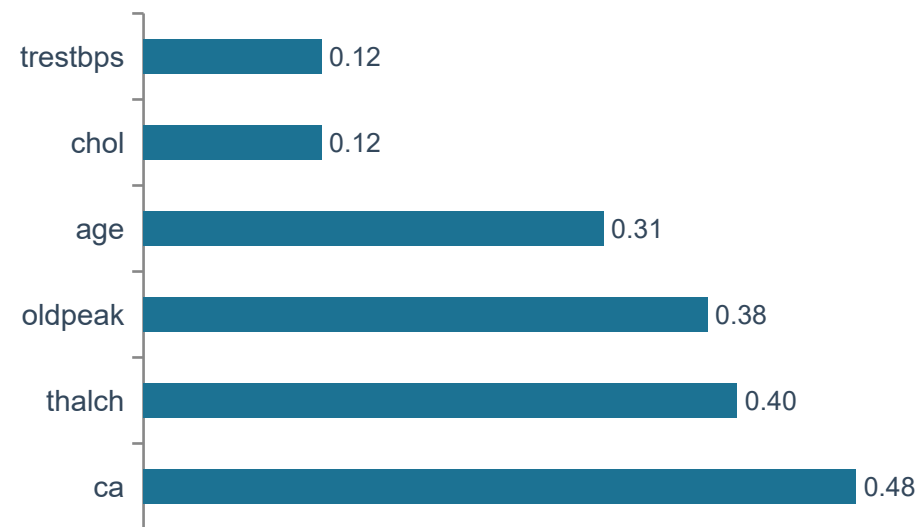
Asymptomatic chest pain shows the HIGHEST disease rate (78%) — the 'silent' group screening can miss.



Feature Correlation with Disease



Strength of link with disease



- ca, thalch, oldpeak, age are the strongest signals.
- chol & BP correlate weakly — famous ≠ predictive.
- No pair exceeds 0.38 → low multicollinearity (good for the baseline).

Baseline Model — Logistic Regression

0.799

Accuracy

0.804

Precision

0.843

Recall

0.744

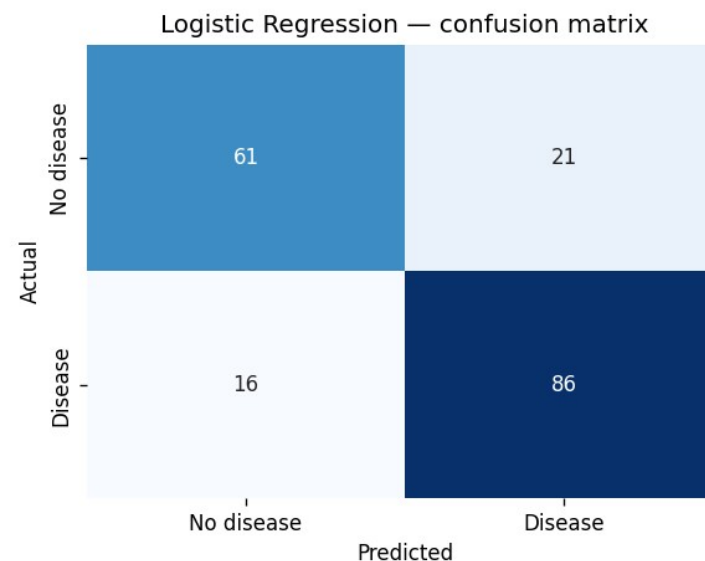
Specificity

0.823

F1

0.897

ROC-AUC



Read it clinically

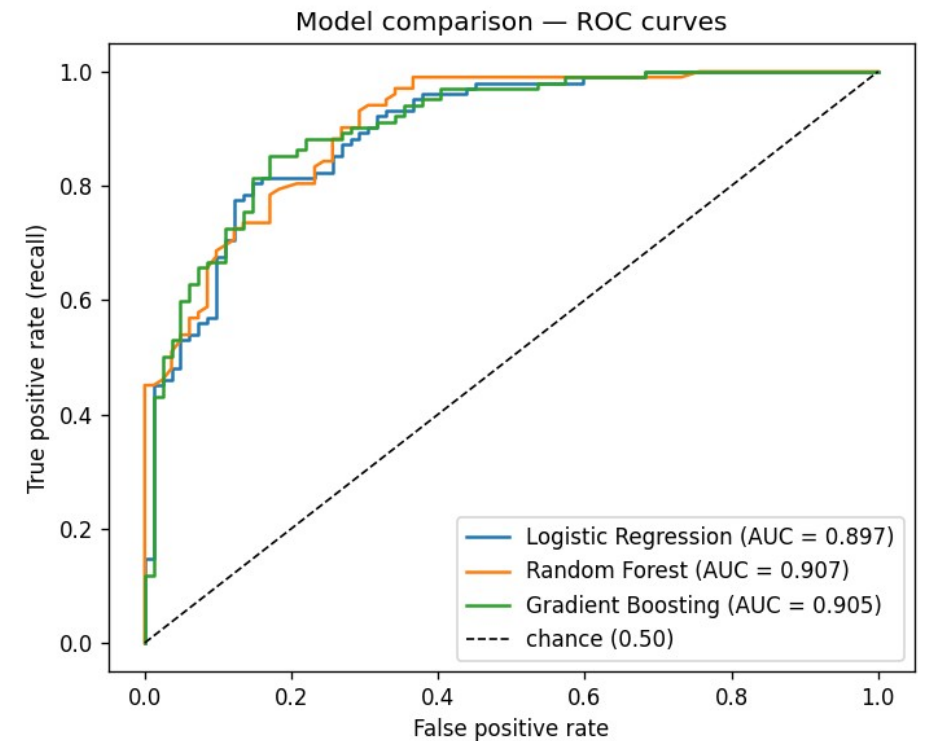
Caught 86 of 102 sick patients; 16 missed cases (false negatives) are the target for tuning. Train 0.836 vs test 0.799 → small gap, a healthy baseline.

Three Models Compared

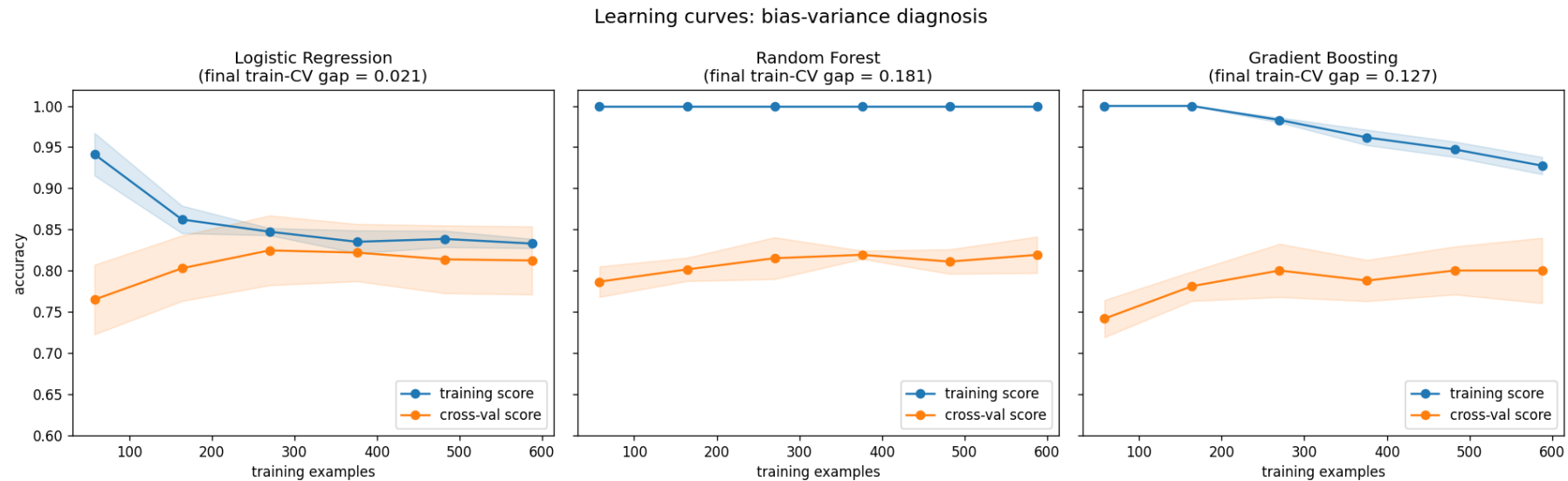
Model	Acc	Recall	Spec	F1	AUC	Train
Logistic Regression	0.799	0.843	0.744	0.823	0.897	0.836
Random Forest	0.810	0.863	0.744	0.834	0.907	1.000
Gradient Boosting ✓	0.821	0.892	0.732	0.847	0.905	0.916

- Both ensembles beat the baseline — real non-linear interactions exist.
- Random Forest hit a perfect 1.000 on training → clear overfitting.
- Winner: Gradient Boosting — best recall (0.892), F1 and accuracy; AUC tied for top.

ROC curves



Bias–Variance Diagnosis (Learning Curves)



Logistic Regression

gap 0.021

Mild bias, low variance — curves meet; more data won't help.

Random Forest

gap 0.181

Severe overfitting — train pinned at 1.000 while CV lags.

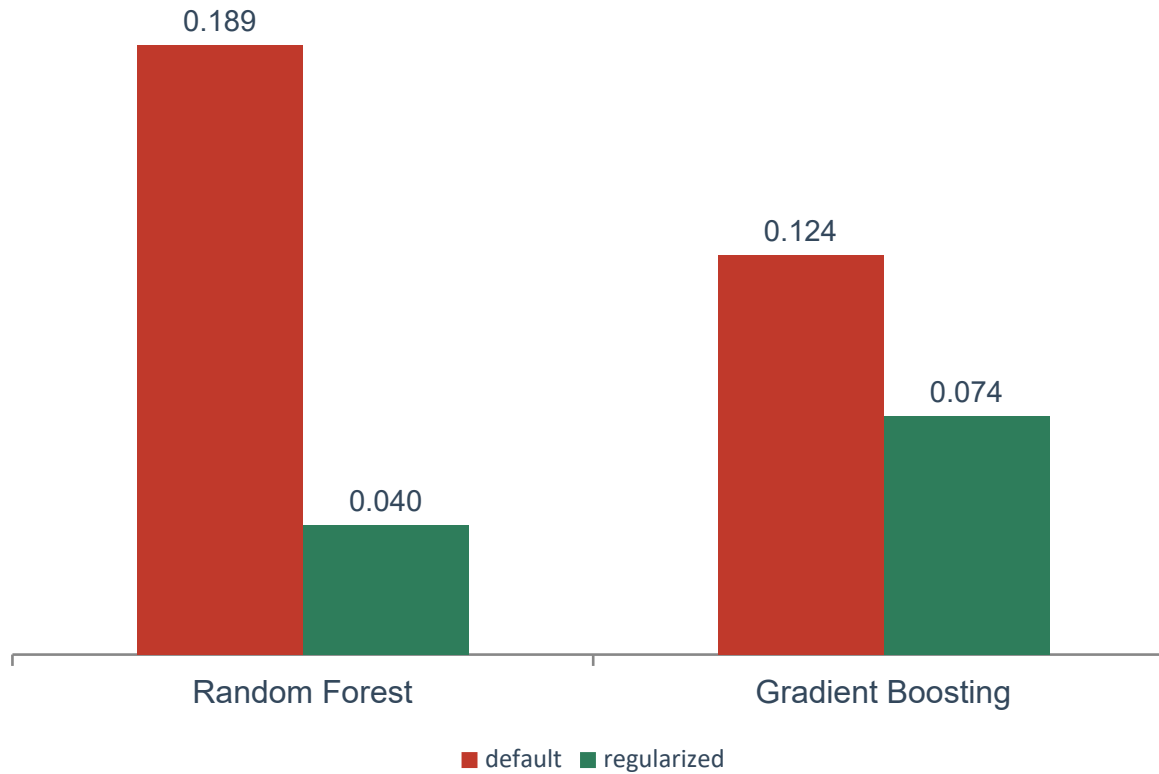
Gradient Boosting

gap 0.127

Moderate, still converging — regularization should help.

Regularization — Shrinking the Variance Gap

Train–CV accuracy gap: before vs after



- Random Forest: gap crushed $0.189 \rightarrow 0.040$ (depth & leaf limits).
- Gradient Boosting: gap halved $0.124 \rightarrow 0.074$ — and CV accuracy rose.
- Logistic Regression is bias-limited: regularization couldn't help it.
- L1 (lasso) drove 11 of 27 coefficients to exactly zero — automatic feature selection.

The lesson

The cure depends on the diagnosis: high variance $\rightarrow$ constrain complexity; high bias $\rightarrow$ add capacity.

Threshold Tuning — Prioritizing Recall

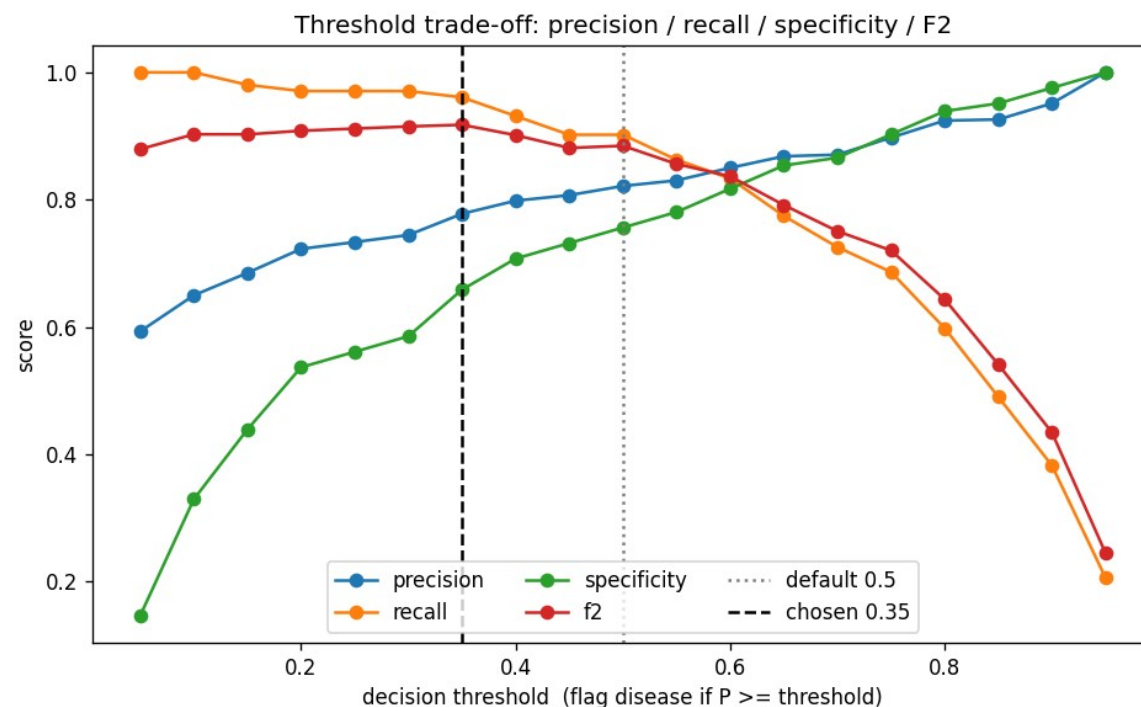
Default 0.50 vs tuned 0.35

Metric	0.50	0.35
Missed cases (FN)	10	4
Recall	0.902	0.961
Precision	0.821	0.778
Specificity	0.756	0.659

-60%
missed cases

Why lower?

A missed heart disease $\gg$ a false alarm. We accept 8 extra alarms to catch 6 more real cases.

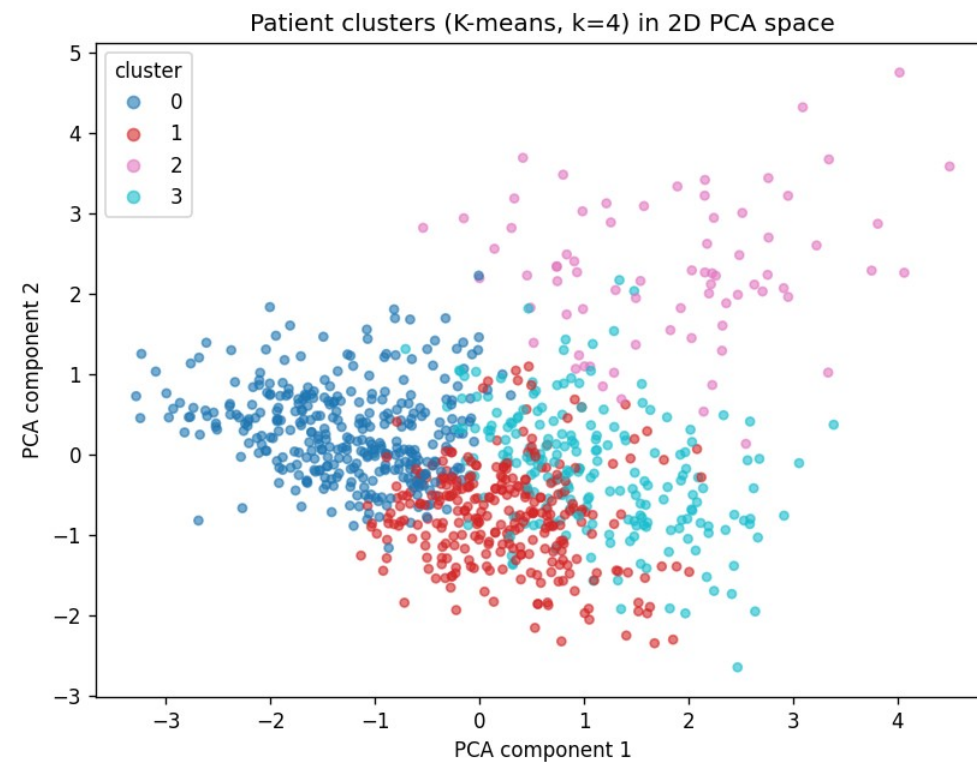


Risk Phenotypes — K-means (k = 4)

Pri	n	Disease	Phenotype
1	74	0.84	Advanced multi-vessel disease
2	211	0.74	Hypertensive-hypercholesterolemic
3	292	0.66	Reduced-exertion / low max-HR
4	343	0.28	Younger lower-risk

- Unsupervised: target hidden during fitting, used only to interpret.
- Chose k=4 for clinical granularity (silhouette was flat) — a transparent judgment call.
- Each phenotype gives the AI agent a priority hint beyond raw probability.

Clusters in 2D (PCA view)



Persist & Bridge to the AI Phase

Saved artifacts (models/)

- risk_model.joblib — regularized Gradient Boosting pipeline (refit on all 920).
- phenotype_model.joblib — K-means impute+scale+cluster.
- metadata.json — threshold 0.35, feature list, phenotype map, test metrics.

0.961

Recall

0.908

ROC-AUC

0.826

Accuracy

Held-out test metrics @ threshold 0.35

```
model_api.assess(patient)
```

HIGH-risk patient

```
probability : 0.976 → high-risk  
phenotype   : Advanced multi-vessel (priority 1)
```

LOW-risk patient

```
probability : 0.013 → cleared  
phenotype   : Younger lower-risk (priority 4)
```

One clean call returns risk + phenotype — the launchpad for Phase B.

Phase A — Headline Results

0.908

ROC-AUC

0.961

Recall (tuned)

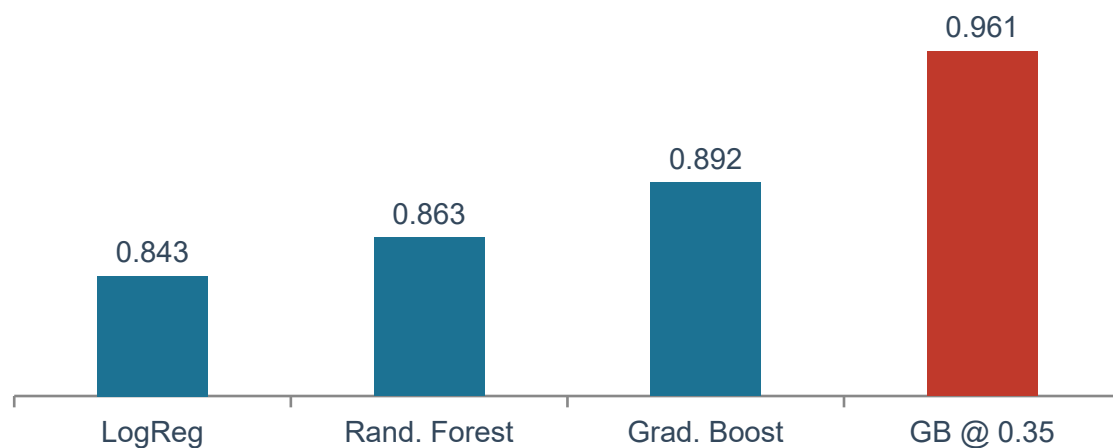
4 / 102

Missed cases

4

Risk phenotypes

Recall across models



What we delivered

- Best model: regularized Gradient Boosting.
- Full leakage-safe pipeline, reused across every step.
- Tuned for safety: only 4 missed cases out of 102.
- Four interpretable phenotypes to guide triage.
- One-call interface bridging ML → AI.

Machine Learning Phase: Complete

A trained, evaluated, deployable cardiac-risk engine.

- Inspected, cleaned, and encoded 920 records — leakage-safe.
- Compared three models; chose Gradient Boosting on recall & AUC.
- Diagnosed bias/variance, regularized, and tuned the threshold for safety.
- Discovered four clinical risk phenotypes with K-means.
- Saved the model and exposed `assess(patient)` for the AI phase.

Next

Say “start the AI phase” to build A* search, the CSP allocator, the knowledge base, and the Gemini + LangChain agent — all on top of `assess()`.